Mathematical Analysis 2

Exercise class 1: a first look at the course

Peter Zeman

This first class comes before the first lecture. We will use ideas from one-variable calculus and linear algebra to see what changes in several variables. The definitions and selected derivations below give a preview; the lectures will develop the theory systematically.

1 The derivative is a linear approximation

Recall the one-variable definition. Let f be defined on an open interval containing a. Its derivative at a can be written in either of two equivalent forms:

$$f'(a) = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h} = \lim_{x\to a} \frac{f(x)-f(a)}{x-a},$$

provided the limit exists. Indeed, substituting $h = x - a$, or $x = a + h$, makes the two quotients identical, and $x \to a$ is equivalent to $h \to 0$. In both limits the denominator is nonzero before taking the limit.

Throughout these notes, we mainly use the first form: a is the fixed point, and h is the increment tending to zero, so $a + h$ approaches a. When viewing the derivative itself as a function, we write $x \mapsto f'(x)$.

Reformulate the definition. An input change h produces the output change $f(a+h) - f(a)$. We seek a fixed $A \in \mathbb{R}$ such that Ah predicts this change with error negligible compared with the step size. By the definition of the derivative,

$$\begin{aligned} \lim_{h\to 0} \frac{f(a+h)-f(a)}{h} = A \iff & \quad \lim_{h\to 0} \left(\frac{f(a+h)-f(a)}{h} - A \right) = 0 \\ \iff & \quad \lim_{h\to 0} \frac{[f(a+h)-f(a)] - Ah}{h} = 0. \end{aligned}$$

The numerator is the **error**: actual change minus predicted change. The last limit says that the error is negligible compared with the step size.

Equivalent definition. The function f is differentiable at a if and only if some fixed $A \in \mathbb{R}$ satisfies

$$f(a+h) - f(a) = Ah + \operatorname{err}(h), \qquad \lim_{h\to 0} \frac{\operatorname{err}(h)}{h} = 0.$$

Then $A = f'(a)$, and the error condition is abbreviated $\operatorname{err}(h) = o(h)$.

Example: the error condition selects A. For $f(x) = x^2$ at $a = 1$, expanding gives $\operatorname{err}(h) = (2 - A)h + h^2$, so $\operatorname{err}(h)/h = 2 - A + h \to 2 - A$. Only $A = 2$ makes this limit zero; then the error is h^2 and $f'(1) = 2$.

Non-example: the error is too large. For $f(x) = \sqrt[3]{x}$ at $a = 0$, any coefficient A gives

$$\operatorname{err}(h) = [f(0+h) - f(0)] - Ah = \sqrt[3]{h} - 0 - Ah = \sqrt[3]{h} - Ah.$$

This error tends to zero, but

$$\frac{\operatorname{err}(h)}{h} = \frac{1}{(\sqrt[3]{h})^2} - A \longrightarrow +\infty.$$

No choice of A makes this quotient tend to zero, so f is not differentiable at zero. For small positive h, the cube root is much larger than h, whereas the quadratic error h^2 in the previous example is much smaller.

What survives in several variables? For $h \in \mathbb{R}^n$ there is no division by the vector h. But a linear approximation still makes sense: replace multiplication by $A \in \mathbb{R}$ with a linear map $A\colon \mathbb{R}^n \to \mathbb{R}^m$. To compare sizes, we divide the norm of the error by the norm of the increment.

Definition: total derivative (or differential)

Let $U \subseteq \mathbb{R}^n$ be open, $a \in U$, and $f\colon U \to \mathbb{R}^m$. We say that f is **differentiable at** a if there is a linear map $A\colon \mathbb{R}^n \to \mathbb{R}^m$ such that

$$\lim_{h\to 0} \frac{\|[f(a+h) - f(a)] - Ah\|}{\|h\|} = 0.$$

We write $Df(a) = A$, so that $f(a+h) - f(a) = Df(a)h + o(\|h\|)$.

Here and below we use Euclidean norms and column vectors. The limit must hold as h tends to zero along *any* path. The map $h \mapsto Df(a)h$ is linear; the approximation $x \mapsto f(a) + Df(a)(x-a)$ is affine.

Example directly from the definition. Let $f(x,y) = (x^2 + y, xy)^\mathsf{T}$. At an arbitrary point (a,b), write $h = (p,q)^\mathsf{T}$. Expanding gives

$$\begin{aligned} f(a+p, b+q) - f(a,b) &= \begin{pmatrix} 2ap + q + p^2 \\ bp + aq + pq \end{pmatrix} \\ &= \begin{pmatrix} 2a & 1 \\ b & a \end{pmatrix} \begin{pmatrix} p \\ q \end{pmatrix} + \begin{pmatrix} p^2 \\ pq \end{pmatrix}. \end{aligned}$$

The matrix collects the coefficients of p, q in the linear terms and defines a candidate linear map A. For $h \neq 0$,

$$\frac{\|[f(a+p,b+q) - f(a,b)] - Ah\|}{\|h\|} = \frac{\sqrt{p^4 + p^2q^2}}{\sqrt{p^2+q^2}} = \frac{|p|\sqrt{p^2+q^2}}{\sqrt{p^2+q^2}} = |p| \longrightarrow 0.$$

Thus f is differentiable at every (a,b), and the matrix above represents $Df(a,b)$.

2 Why the derivative matrix contains partial derivatives

Write $f = (f_1, \ldots, f_m)^\mathsf{T}$. Suppose f is differentiable at a, and identify $A = Df(a)$ with its matrix $(a_{ij}) \in \mathbb{R}^{m\times n}$ in the standard bases. Denote its ith row by A_{i*}, so $(Ah)_i = A_{i*}h$. A vector tends to zero if and only if each of its components does. Thus the defining limit can be written componentwise: for every $i = 1, \ldots, m$,

$$\lim_{h\to 0} \frac{f_i(a+h) - f_i(a) - A_{i*}h}{\|h\|} = 0.$$

Now take $h = te_j$, where e_j is the jth standard basis vector. Only the jth input coordinate changes, and $A_{i*}h = ta_{ij}$. For $t \neq 0$, the ratio $\|te_j\|/t$ is either 1 or -1, so replacing $\|h\|$ by t in the denominator preserves the zero limit. Hence

$$\lim_{t\to 0} \frac{f_i(a+te_j) - f_i(a) - a_{ij}t}{t} = 0.$$

Rearranging gives

$$a_{ij} = \lim_{t\to 0} \frac{f_i(a+te_j) - f_i(a)}{t} = \frac{\partial f_i}{\partial x_j}(a).$$

This is an ordinary derivative with only the jth input coordinate varying; the other coordinates are fixed. It is called the **partial derivative** of f_i with respect to x_j. Thus the defining limit forces every matrix entry a_{ij} to equal the corresponding partial derivative.

The Jacobian matrix

In the standard bases, the matrix of $Df(a)$ is

$$J_f(a) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(a) & \cdots & \frac{\partial f_1}{\partial x_n}(a) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(a) & \cdots & \frac{\partial f_m}{\partial x_n}(a) \end{pmatrix} \in \mathbb{R}^{m\times n}, \qquad Df(a)h = J_f(a)h.$$

Rows correspond to output components; columns to input coordinates.

The same example using the Jacobian. For $f(x,y) = (x^2+y, xy)^\mathsf{T}$, computing the first partial derivatives gives

$$J_f(a,b) = \begin{pmatrix} 2a & 1 \\ b & a \end{pmatrix}, \qquad J_f(1,1) = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

This is exactly the matrix obtained earlier directly from the definition.

A distinction to remember. Existence of the partial derivatives alone does not imply differentiability. For instance, $g(x,y) = xy/(x^2+y^2)$ away from $(0,0)$, with $g(0,0) = 0$, has both partial derivatives zero at the origin, but $g(t,t) = 1/2$ for $t \neq 0$: it is not even continuous there. A useful sufficient condition is that all first partial derivatives exist and are continuous in a neighbourhood of a; then f is differentiable at a. This is the usual C^1 condition.

3 The chain rule: compose the linear approximations

One variable. If f is differentiable at a and g at $f(a)$, then

$$(g\circ f)'(a) = g'(f(a))\, f'(a).$$

For example, $\frac{\mathrm{d}}{\mathrm{d}x}\sin(x^2) = 2x\cos(x^2)$. The derivative of the outer function is evaluated at the value of the inner function.

Chain rule in several variables

Let $U \subseteq \mathbb{R}^n$ and $V \subseteq \mathbb{R}^m$ be open, with $f\colon U \to V$ and $g\colon V \to \mathbb{R}^k$. If f is differentiable at a and g is differentiable at $f(a)$, then

$$D(g\circ f)(a) = Dg(f(a)) \circ Df(a).$$

In matrix notation,

$$\underbrace{J_{g\circ f}(a)}_{k\times n} = \underbrace{J_g(f(a))}_{k\times m}\underbrace{J_f(a)}_{m\times n}.$$

Why this is the right formula. Write $\Delta y = f(a+h) - f(a)$ for the change in the output of f. To first order, $\Delta y \approx Df(a)h$. This is the input change for g, whose input moves from $f(a)$ to $f(a)+\Delta y$. Hence

$$\begin{aligned}(g\circ f)(a+h) - (g\circ f)(a) &= g(f(a)+\Delta y) - g(f(a)) \\ &\approx Dg(f(a))\,\Delta y \\ &\approx Dg(f(a))(Df(a)h).\end{aligned}$$

We first apply $Df(a)$ and then $Dg(f(a))$: the resulting linear map is their composition. This sketches the reasoning; the full proof checks that the combined error is negligible compared with $\|h\|$.

Worked example. Use $f(x,y) = (x^2+y, xy)^\mathsf{T}$ and $g(u,v) = u^2+v^2$. Since $J_g(u,v) = (2u \;\; 2v)$,

$$\begin{aligned} J_{g\circ f}(x,y) &= \begin{pmatrix} 2(x^2+y) & 2xy \end{pmatrix} \begin{pmatrix} 2x & 1 \\ y & x \end{pmatrix} \\ &= \begin{pmatrix} 4x(x^2+y) + 2xy^2 & 2(x^2+y) + 2x^2y \end{pmatrix}. \end{aligned}$$

At $(1,1)$ this is $(10 \;\; 6)$. Directly differentiating $(g\circ f)(x,y) = (x^2+y)^2 + x^2y^2$ gives the same answer.

Scalar functions and curves. For $F\colon \mathbb{R}^n \to \mathbb{R}$, we define its **gradient** as a column vector, consistent with our convention for vectors in $\mathbb{R}^n$:

$$\nabla F(a) = \begin{pmatrix} \partial_1 F(a) \\ \vdots \\ \partial_n F(a) \end{pmatrix}, \qquad J_F(a) = \begin{pmatrix} \partial_1 F(a) & \cdots & \partial_n F(a) \end{pmatrix} = \nabla F(a)^\mathsf{T}.$$

The gradient is $n \times 1$; the matrix of the derivative $DF(a)\colon \mathbb{R}^n \to \mathbb{R}$ is $1 \times n$. Thus $DF(a)h = \nabla F(a)^\mathsf{T} h = \nabla F(a) \cdot h$. Along a differentiable curve $\gamma\colon \mathbb{R} \to \mathbb{R}^n$, the chain rule reads

$$\frac{\mathrm{d}}{\mathrm{d}t} F(\gamma(t)) = \nabla F(\gamma(t)) \cdot \gamma'(t).$$

In particular, the directional derivative along v is $DF(a)v$.

Worked example. Let $F(x,y) = x^2+y^2$ and $\gamma(t) = (\cos t, \sin t)^\mathsf{T}$. The chain rule gives $\frac{\mathrm{d}}{\mathrm{d}t} F(\gamma(t)) = \nabla F(\gamma(t)) \cdot \gamma'(t) = -2\cos t \sin t + 2 \sin t \cos t = 0$. Alternatively, $F(\gamma(t)) = \cos^2 t + \sin^2 t = 1$, whose derivative is zero.

4 Extrema in one variable: go to the first nonzero term

A point a is a **local minimum** if $f(x) \geq f(a)$ for all x sufficiently close to a in the domain. It is **strict** if the inequality is strict for $x \neq a$. Reverse the inequalities for a maximum.

Step 1: find candidates. At an interior local extremum where f is differentiable, $f'(a) = 0$ (Fermat's theorem). Indeed, the difference quotients at a minimum are nonnegative from the right and nonpositive from the left, so their common limit is zero. Such points are called **stationary points**. In the usual search for critical points, also include interior points where the derivative does not exist. For absolute extrema on a closed interval, check the endpoints and compare all candidate values. A continuous function on a closed bounded interval attains both an absolute minimum and maximum.

Step 2: the first derivative test. Assume f is continuous at a and differentiable on either side nearby. The signs of f' describe the behaviour as we pass through a:

$$- \longrightarrow + : \text{ strict local minimum}, \qquad + \longrightarrow - : \text{ strict local maximum}.$$

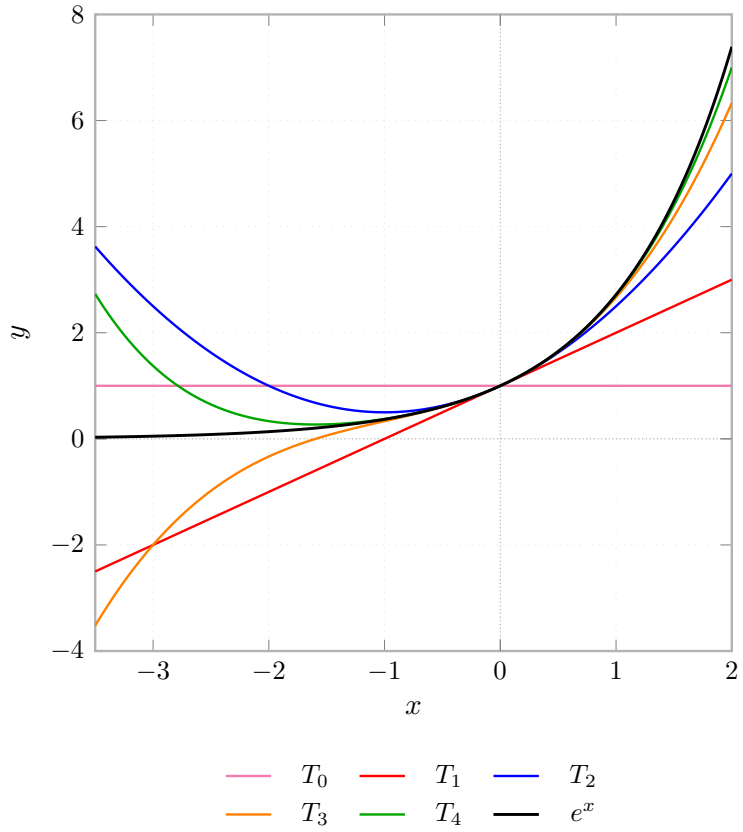
If f' is strictly positive on both sides, or strictly negative on both sides, there is no local extremum. For example, x^3 has $f'(0) = 0$ but is increasing through zero.

Step 3: the second derivative test. If $f \in C^2$ near a and $f'(a) = 0$, Taylor's formula gives

$$f(a+h) - f(a) = \tfrac{1}{2} f''(a) h^2 + o(h^2).$$

Since $h^2 > 0$ for $h \neq 0$, $f''(a) > 0$ gives a strict local minimum, and $f''(a) < 0$ a strict local maximum. If $f''(a) = 0$, this test is inconclusive: x^4, $-x^4$, and x^3 exhibit all three possibilities.

More generally, the Taylor polynomial is $T_k(a+h) = \sum_{j=0}^{k} f^{(j)}(a) h^j / j!$.



Taylor polynomials for e^x at 0: $T_k(x) = \sum_{j=0}^{k} x^j/j!$. All derivatives at 0 equal 1, so both even and odd powers occur, and every degree gives a different polynomial.

For extrema, we look for the first nonconstant term with a nonzero coefficient.

Higher derivative test via Taylor's polynomial

Suppose $f \in C^k$ near a (its first k derivatives are continuous), where $k \geq 2$, and

$$f'(a) = \cdots = f^{(k-1)}(a) = 0, \qquad f^{(k)}(a) \neq 0.$$

Then the first nonconstant term decides the sign near a:

$$f(a+h) - f(a) = ch^k + o(h^k), \qquad c = \frac{f^{(k)}(a)}{k!}.$$

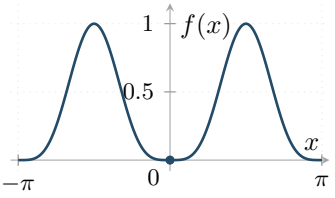
Order k	Coefficient	Conclusion at a
Even	$c > 0$	Strict local minimum
Even	$c < 0$	Strict local maximum
Odd	$c \neq 0$	No local extremum

Examples at zero. $x^4+x^5 = x^4(1+x)$ has a strict local minimum; $-x^6$ has a strict local maximum; x^5 has neither.

A nonpolynomial example is $f(x) = \sin^4 x$. Direct differentiation gives

$$f'(x) = 4\sin^3 x \cos x, \qquad f''(x) = 12\sin^2 x\cos^2 x - 4\sin^4 x,$$
$$f'''(x) = 24\sin x\cos^3 x - 40\sin^3 x\cos x, \quad f^{(4)}(x) = 24\cos^4 x - 192\sin^2 x\cos^2 x + 40\sin^4 x.$$

At 0, $f'(0) = f''(0) = f'''(0) = 0$, while $f^{(4)}(0) = 24 > 0$. Thus $f(x) = x^4 + o(x^4)$, and 0 is a strict local minimum.



If all derivatives vanish, there may be no first nonzero term. For example, $f(0) = 0$, $f(x) = e^{-1/x^2}$ for $x \neq 0$ is smooth with all derivatives zero at 0, yet has a strict minimum there.

5 Several variables: quadratic forms replace quadratic terms

Let $f\colon U \subseteq \mathbb{R}^n \to \mathbb{R}$ be differentiable and $a \in U$ an interior local extremum. Restricting f to each line $t \mapsto a + tv$ gives $Df(a)v = 0$ for every v. Hence

$$Df(a) = 0 \qquad \Longleftrightarrow \qquad \nabla f(a) = 0.$$

We again start by finding stationary points. Boundary extrema require separate analysis.

Quadratic terms in several variables. In one variable, the quadratic part of a Taylor polynomial is ch^2. In several variables, terms of total degree two are products h_ih_j, including mixed terms: for example, $\alpha h_1^2 + 2\beta h_1h_2 + \gamma h_2^2$ in two variables. These are called **quadratic forms** and can be written as $q(h) = h^{\mathsf{T}}Ah$ with A symmetric. Along each line $h = tv$, we have $q(tv) = q(v)t^2$: a one-variable quadratic term whose coefficient depends on the direction v.

If $f \in C^2$ near a, define its **Hessian matrix** by

$$H_f(a) = \left(\partial_i\partial_j f(a)\right)_{i,j=1}^{n}.$$

It is symmetric because continuous mixed second partial derivatives agree. The second-order Taylor formula is

$$f(a+h) = f(a) + \nabla f(a)\cdot h + \tfrac{1}{2}h^{\mathsf{T}}H_f(a)h + o(\|h\|^2).$$

At a stationary point, the linear term disappears. The role of $f''(a)h^2$ is played by the **quadratic form** $q(h) = h^{\mathsf{T}}H_f(a)h$.

Hessian test at a stationary point

Sign of $q(h)$	**Conclusion**
$q(h) > 0$ for every $h \neq 0$ (positive definite)	Strict local minimum
$q(h) < 0$ for every $h \neq 0$ (negative definite)	Strict local maximum
q takes both positive and negative values (indefinite)	Saddle point; no local extremum
$q \geq 0$ or $q \leq 0$, but $q(h) = 0$ for some $h \neq 0$	Test inconclusive

Connection with linear algebra. In orthonormal eigenvector coordinates, $q(h) = \lambda_1 z_1^2 + \cdots + \lambda_n z_n^2$. All positive eigenvalues give a minimum, all negative a maximum, and both signs give a saddle. In the positive definite case, $q(h) \geq \lambda_{\min}\|h\|^2$ with $\lambda_{\min} > 0$; this uniform bound lets the quadratic term dominate the remainder.

Examples at $(0,0)$. The functions x^2+y^2, $-x^2-y^2$, and x^2-y^2 give a minimum, a maximum, and a saddle, respectively. In the last case the x-axis gives positive values and the y-axis negative values. The functions x^2+y^4 and x^2-y^4 have the same Hessian $\operatorname{diag}(2,0)$ at the origin, but the first has a strict minimum and the second a saddle. A zero eigenvalue need not prevent a conclusion if other eigenvalues already have both signs.

A practical test in two variables. For $H = \begin{pmatrix} A & B \\ B & C \end{pmatrix}$, let $\Delta = AC - B^2$. If $\Delta > 0$, the sign of A distinguishes a minimum from a maximum; if $\Delta < 0$, there is a saddle; if $\Delta = 0$, the test is inconclusive.

Worked example. For $f(x,y) = x^2 + 2xy + 3y^2$, we have $f(0,0) = 0$. Completing the square gives $f(x,y) = (x+y)^2 + 2y^2 > 0$ away from the origin, so the origin is a strict minimum.

6 Integration: add up small contributions

Recall the integral via Darboux sums. Let $f\colon [a,b] \to \mathbb{R}$ be bounded. For a partition $P : a = x_0 < x_1 < \cdots < x_N = b$, set $m_i = \inf_{[x_{i-1},x_i]} f$ and $M_i = \sup_{[x_{i-1},x_i]} f$. The **lower and upper sums** are

$$L(f,P) = \sum_{i=1}^{N} m_i \Delta x_i, \qquad U(f,P) = \sum_{i=1}^{N} M_i \Delta x_i, \qquad \Delta x_i = x_i - x_{i-1}.$$

The function is **Riemann integrable** when its lower and upper integrals agree:

$$\int_a^b f(x)\,\mathrm{d}x = \sup_P L(f,P) = \inf_P U(f,P).$$

For $f \geq 0$, these sums bound the area under the graph from below and above using rectangles. With both signs present, the integral is signed area. The fundamental theorem of calculus supplies a way to calculate it: if f is continuous and $F' = f$, then $\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a)$.

The multivariable version. For a bounded function on a box $B \subseteq \mathbb{R}^n$, partition B into subboxes Q_α. Replace interval length by n-dimensional volume:

$$L(f,P) = \sum_\alpha (\inf_{Q_\alpha} f)\,\mathrm{Vol}_n(Q_\alpha), \qquad U(f,P) = \sum_\alpha (\sup_{Q_\alpha} f)\,\mathrm{Vol}_n(Q_\alpha).$$

Here $\mathrm{Vol}_n(\prod_{j=1}^n [a_j,b_j]) = \prod_{j=1}^n (b_j - a_j)$. Again, integrability means $\sup_P L(f,P) = \inf_P U(f,P)$; this common value defines $\int_B f(x)\,\mathrm{d}x$. In two dimensions, $\mathrm{d}x\,\mathrm{d}y$ represents area; in three, $\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$ represents volume. In particular,

$$\mathrm{Area}(D) = \iint_D 1\,\mathrm{d}x\,\mathrm{d}y, \qquad \mathrm{Vol}_3(E) = \iiint_E 1\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z.$$

Fubini's theorem: a useful introductory version

If f is continuous on $[a,b] \times [c,d]$, then

$$\begin{aligned} \iint_{[a,b]\times[c,d]} f(x,y)\,\mathrm{d}x\,\mathrm{d}y &= \int_a^b \left(\int_c^d f(x,y)\,\mathrm{d}y \right) \mathrm{d}x \\ &= \int_c^d \left(\int_a^b f(x,y)\,\mathrm{d}x \right) \mathrm{d}y. \end{aligned}$$

Thus either variable may be integrated first. The analogous statement holds in higher dimensions.

For a bounded region D, enclose it in a box and extend f by zero outside D. If this extension is Riemann integrable, its integral defines $\int_D f$. The disks and regions bounded by continuous graphs used here are Jordan measurable: their boundaries can be covered by finitely many boxes of arbitrarily small total volume.

For a region $D = \{(x, y) : a \le x \le b,\ \alpha(x) \le y \le \beta(x)\}$ with continuous bounds $\alpha \le \beta$ and continuous f, the slicing version is

$$\iint_D f(x, y)\,\mathrm{d}x\,\mathrm{d}y = \int_a^b \int_{\alpha(x)}^{\beta(x)} f(x, y)\,\mathrm{d}y\,\mathrm{d}x.$$

Changing the order requires describing the *same region* with new limits. For improper or singular integrals, further hypotheses are needed; the order cannot be interchanged without checking them.

7 Area of a disk I: Cartesian slices and Fubini

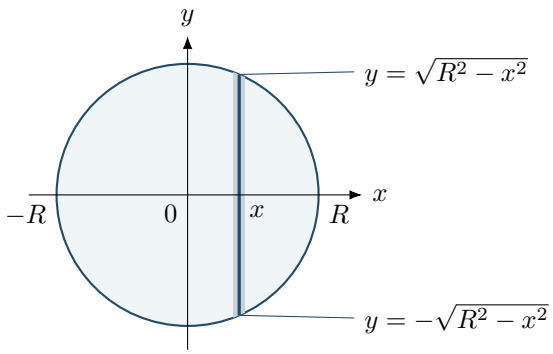
Let $R > 0$ and let

$$D_R = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \le R^2\}.$$

We want the area enclosed by the circle, that is, the area of the disk: $\mathrm{Area}(D_R) = \iint_{D_R} 1\,\mathrm{d}x\,\mathrm{d}y$.

Find the limits from the inequality. A vertical slice exists exactly when $-R \le x \le R$. For such a fixed x,

$$x^2 + y^2 \le R^2 \quad \Longleftrightarrow \quad -\sqrt{R^2 - x^2} \le y \le \sqrt{R^2 - x^2}.$$



Integrate first along each slice. Fubini gives

$$\begin{aligned}\mathrm{Area}(D_R) &= \int_{-R}^{R} \int_{-\sqrt{R^2-x^2}}^{\sqrt{R^2-x^2}} 1\,\mathrm{d}y\,\mathrm{d}x \\ &= \int_{-R}^{R} 2\sqrt{R^2 - x^2}\,\mathrm{d}x.\end{aligned}$$

The inner integral is the length of a vertical slice. The outer integral adds these lengths, weighted by the horizontal width $\mathrm{d}x$.

Evaluate using one-variable substitution. Put $x = R\sin t$, so $\mathrm{d}x = R\cos t\,\mathrm{d}t$ and $-\pi/2 \le t \le \pi/2$. On this interval $\cos t \ge 0$, hence $\sqrt{R^2 - x^2} = R\cos t$. Therefore

$$\begin{aligned}\mathrm{Area}(D_R) &= 2R^2 \int_{-\pi/2}^{\pi/2} \cos^2 t\,\mathrm{d}t \\ &= R^2 \left[t + \tfrac{1}{2}\sin(2t)\right]_{-\pi/2}^{\pi/2} = \boxed{\pi R^2}.\end{aligned}$$

Worked example. To integrate with respect to x first, use horizontal slices:

$$\operatorname{Area}(D_R) = \int_{-R}^{R} \int_{-\sqrt{R^2-y^2}}^{\sqrt{R^2-y^2}} 1 \,\mathrm{d}x \,\mathrm{d}y = \int_{-R}^{R} 2\sqrt{R^2-y^2} \,\mathrm{d}y = \pi R^2.$$

The resulting one-variable integral is the same as before, with y in place of x.

8 Area of a disk II: change of variables and the determinant

Recall one-variable substitution. For continuous f and a C^1 map ϕ into its interval of definition,

$$\int_{\phi(\alpha)}^{\phi(\beta)} f(x) \,\mathrm{d}x = \int_{\alpha}^{\beta} f(\phi(t))\phi'(t) \,\mathrm{d}t.$$

This formula uses oriented interval limits. For integration over geometric regions, the local length factor is $|\phi'(t)|$. In n dimensions the corresponding volume factor is an absolute determinant.

Change of variables: the formula to remember

A C^1 **diffeomorphism** $\Phi\colon U \to V$ between open subsets of $\mathbb{R}^n$ is a bijection such that both Φ and Φ^{-1} are C^1. For a compact Jordan measurable $E \subset U$ and continuous f on $\Phi(E)$,

$$\int_{\Phi(E)} f(x) \,\mathrm{d}x = \int_{E} f(\Phi(u)) \,\left|\det J_\Phi(u)\right| \,\mathrm{d}u.$$

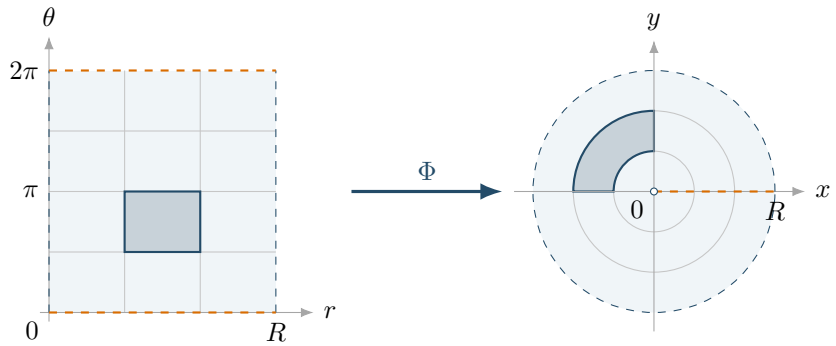
Versions with boundary limits extend this formula to the polar calculation below.

Why a determinant? Recall from linear algebra that a linear map with matrix $A \in \mathbb{R}^{n\times n}$ multiplies volume by $|\det A|$; the sign of $\det A$ records orientation. Near u, Φ is approximated by its derivative $D\Phi(u)$, whose matrix is $J_\Phi(u)$. This explains the local volume factor $|\det J_\Phi(u)|$.

Polar coordinates. Set $\Phi(r,\theta) = (r\cos\theta, r\sin\theta)$. Then

$$J_\Phi(r,\theta) = \begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix}, \qquad \det J_\Phi(r,\theta) = r.$$

Thus $\mathrm{d}x\,\mathrm{d}y = r\,\mathrm{d}r\,\mathrm{d}\theta$. A small polar cell has radial thickness Δr and tangential length approximately $r\Delta\theta$. Its area is approximately $r\Delta r\Delta\theta$, explaining the same factor geometrically.



Parameter rectangle Disk with a slit

The highlighted rectangle maps to the highlighted sector. The orange edges correspond to the omitted radius.

Where is this a diffeomorphism? Restrict to $0 < r < R$ and $0 < \theta < 2\pi$. This maps diffeomorphically onto the open disk with the nonnegative x-axis removed. We exclude $r = 0$ because the determinant vanishes there, and $\theta = 0, 2\pi$ because these angles describe the same point for each fixed r. The omitted radius and boundary circle have area zero and do not change the integral of 1. One can justify the boundary limits by first using compact subrectangles inside the parameter domain and then taking their limits.

The disk condition becomes simply $0 \le r \le R$, with one full turn $0 \le \theta \le 2\pi$. Therefore

$$\operatorname{Area}(D_R) = \int_0^{2\pi} \int_0^R 1 \cdot r \, \mathrm{d}r \, \mathrm{d}\theta = \int_0^{2\pi} \frac{R^2}{2} \, \mathrm{d}\theta = \boxed{\pi R^2}.$$

The factor r is essential: a rectangle in (r, θ) coordinates does not have the same area as its image in the plane.

9 Why metric spaces belong in this course

We have repeatedly used phrases such as "$h \to 0$", "near a", and "continuous". They require a way to measure distance. Metric spaces isolate the structure needed to discuss limits, continuity, and compactness, including on sets whose elements are not vectors.

Start with linear algebra. On a real inner product space, define

$$\|v\| = \sqrt{\langle v, v \rangle}, \qquad d(x, y) = \|x - y\| .$$

For the standard inner product on $\mathbb{R}^n$, these become

$$\langle x, y \rangle = \sum_{j=1}^n x_j y_j, \qquad \|x\|_2 = \sqrt{\sum_{j=1}^n x_j^2}, \qquad d_2(x, y) = \sqrt{\sum_{j=1}^n (x_j - y_j)^2}.$$

A **norm** on a real vector space is a function into $[0, \infty)$ satisfying

$$\|v\| = 0 \iff v = 0, \qquad \|\lambda v\| = |\lambda| \, \|v\| , \qquad \|u + v\| \le \|u\| + \|v\| .$$

For an inner product, the triangle inequality follows from Cauchy–Schwarz. Every norm then induces a distance because, for example,

$$d(x, z) = \|(x - y) + (y - z)\| \le \|x - y\| + \|y - z\| .$$

Definition: metric space

A metric on a nonempty set X is a function $d \colon X \times X \to [0, \infty)$ such that, for all $x, y, z \in X$,

$$\begin{aligned} d(x, y) = 0 \quad &\iff \quad x = y, \\ d(x, y) &= d(y, x), \\ d(x, z) &\le d(x, y) + d(y, z). \end{aligned}$$

The pair (X, d) is a **metric space**. No addition of points, scalar multiplication, or angles are required.

The implications are

$$\text{inner product} \implies \text{norm} \implies \text{metric}.$$

The reverse implications need not hold. On $\mathbb{R}^2$, $\|(x, y)\|_1 = |x| + |y|$ and $\|(x, y)\|_\infty = \max\{|x| , |y|\}$ are norms, but are not induced by inner products. Their unit balls are a diamond and a square.

On any nonempty set, the **discrete metric** $d(x, y) = 1$ for $x \neq y$, and $d(x, x) = 0$, needs no vector space at all.

The familiar definitions now have one common form. Convergence means $x_k \to x$ if $d(x_k, x) \to 0$. An open ball is $B_d(a, r) = \{x : d(x, a) < r\}$, and a set is open if it contains a ball about each of its points. A map $F\colon (X, d_X) \to (Y, d_Y)$ is continuous at a if for every $\varepsilon > 0$ there is $\delta > 0$ such that

$$d_X(x, a) < \delta \quad \Longrightarrow \quad d_Y(F(x), F(a)) < \varepsilon.$$

For example, $d(f, g) = \max_{t\in[0,1]} |f(t) - g(t)|$ is a metric on $C([0, 1])$, the space of continuous functions. Convergence in this metric is uniform convergence: functions themselves can be points of a space. For derivatives, we additionally need the linear structure used in $a + h$ and $Df(a)h$; a metric alone is not enough.